Internship report

Development of a COMSOL Breast FEM Model for the Early Warning Scan

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1 Abstract

This internship project developed a reproducible COMSOL finite-element workflow for modelling dynamic breast deformation in the context of the Early Warning Scan. The work focused on improving a parametrised breast model by adding a curved posterior support, a more realistic fibroglandular tissue distribution, volumetric skin representation, material-sensitivity routes, simplified Cooper-like support and analytic tumor-overlay placement.

The workflow generated traceable COMSOL model variants and compact post-processing outputs for comparing selected modelling choices. Material and skin-parameter changes produced the strongest displacement differences in the available result sets, while the simplified Cooper-like support variants produced only small global displacement changes. The tumor-overlay route was implemented and visualised at build level for different tumor sizes and locations, but the current results should not be interpreted as evidence of tumor detectability.

The main limitation was computational capacity, because time-dependent simulations and post-processing were performed on local hardware. The developed workflow therefore provides a model-development platform for future EWS-oriented deformation studies, but further work should use stronger computational resources, larger matched simulation sets and experimental or EWS-like validation data.

2 Introduction

Breast cancer is the most commonly diagnosed cancer among women worldwide, with more than 2.3 million new cases reported annually [1]. Early detection is essential for improving treatment options and patient outcome. X-ray mammography is currently the standard method for population-based breast cancer screening, but it also has relevant limitations. In particular, mammographic sensitivity is reduced in women with dense breast tissue, where the contrast between lesions and surrounding fibroglandular tissue can be limited [2, 3]. In addition, mammography uses ionizing radiation and requires breast compression, which motivates continued research into complementary, non-invasive and radiation-free screening approaches.

2.1 Background

One possible direction for radiation-free breast cancer detection is to use the mechanical contrast between healthy and pathological tissue. Malignant breast tumours are generally stiffer than surrounding adipose and fibroglandular tissue, although reported stiffness values vary with measurement technique, loading condition and tissue type [4–6]. This mechanical contrast forms the basis for elastography-based imaging techniques, in which tissue deformation is used to infer differences in internal mechanical properties [7–9].

This project was performed in collaboration with Early Warning Scan (EWS), which is developing a non-invasive screening concept based on optical and biomechanical measurements of breast deformation [10]. The underlying principle is related to Digital Image Elasto-Tomography (DIET), where the breast is mechanically excited and the resulting surface motion is captured using optical measurements. Differences in internal stiffness may then affect the measured surface displacement field and dynamic response [11–13]. Previous phantom and modelling studies have shown that finite element (FE) models can support the development of DIET-like systems by generating controlled deformation data and by testing the effect of stiff inclusions under known conditions [14, 15].

Finite element method (FEM) modelling is widely used to study breast biomechanics because it allows geometry, material properties, boundary conditions and loading conditions to be controlled independently. Early breast FEM studies already demonstrated the usefulness of three-dimensional biomechanical models for simulating breast deformation and image-related applications [16, 17]. More recent work has moved toward patient-specific and multi-component models that include separate anatomical regions such as adipose tissue, fibroglandular tissue, skin, muscle and support structures, each with distinct mechanical properties [16–19]. Such models can be used to study breast motion, compression, gravity-induced deformation, image registration and the effect of internal tissue composition on the external deformation response.

However, breast modelling remains challenging. Breast tissue is heterogeneous, nonlinear and individual-specific, and its mechanical properties vary substantially between studies, measurement protocols and anatomical regions [5, 6, 20, 21]. Anatomical features such as the posterior chestwall interface, fibroglandular distribution, skin layer and Cooper-ligament support can all influence the simulated deformation response. At the same time, adding anatomical detail increases model complexity, solver cost and the risk of poorly controlled comparisons. For an EWS-oriented modelling workflow, it is therefore important not only to make the model more realistic, but also to keep the development process reproducible and interpretable.

2.2 Objectives

The aim of this internship project was to develop a reproducible COMSOL-based FEM pipeline for a parametrised breast model and to use this pipeline to investigate how selected anatomical and mechanical modelling choices influence the simulated deformation response. The project started from the objective of improving the realism of an existing breast model, with an initial emphasis on geometry, posterior chestwall support and internal tissue representation. The focus was not to create a final patient-specific diagnostic model, but to establish a controlled model-development workflow in which model variants can be generated, solved and compared systematically.

The central research question of this project was:

> How do controlled changes in chestwall geometry, glandular tissue distribution, outer breast shape, support representation, material settings and tumor inclusion affect the simulated displacement and stress response of a heterogeneous finite element breast model?

To address this question, the model was developed in a staged manner. The first stages focused on establishing a stable dynamic baseline and improving the anatomical description of the model through the posterior chestwall geometry and glandular tissue distribution. Subsequent stages introduced matched controls for outer-shape asymmetry and Cooper-ligament support, followed by material and skin-layer sensitivity cases and an initial tumor screening route. This staged structure was chosen to make the effect of individual modelling assumptions easier to interpret.

COMSOL Multiphysics was used as the intended modelling environment for this project, but an important part of the work was to determine whether a sufficiently automated and reproducible COMSOL workflow could be built for this type of parametrised breast model [22]. The resulting pipeline supports systematic comparison between model variants while keeping the model definitions and evaluation outputs traceable. The final outcome of the project is therefore a reproducible COMSOL FEM model-development pipeline for EWS-oriented breast deformation studies, together with a critical assessment of which model components are suitable for quantitative interpretation and which should still be considered exploratory model-development results.

3 Materials and Methods

3.1 Model-development workflow

3.1.1 Starting point and relation to previous EWS FEM work

This project was developed in the context of previous internal EWS finite element modelling work. Earlier EWS models provided a useful methodological starting point for breast geometry parameterisation, soft-tissue material assumptions, dynamic loading and displacement-based evaluation [23, 24]. These previous projects were used as engineering reference material and as a comparison point for modelling choices. They were not used as validation datasets for the current COMSOL model.

The present project develops a separate COMSOL-based model. This was done to improve control over constructive geometry, region definitions, model variants and post-processing while keeping the workflow reproducible. This was especially relevant because the project aimed to investigate the influence of individual anatomical and mechanical components, including chestwall shape, fibroglandular tissue distribution, skin representation, Cooper-ligament support approximation and tumor inclusion. Additional context on the previous EWS FEM work is provided in Appendix A, while the current repository structure is summarised in Appendix B.

3.1.2 COMSOL-based workflow

The finite element model was implemented in COMSOL Multiphysics and driven by a Python-based source-case workflow [22]. Each model variant was defined by a TOML case file, where TOML refers to a text configuration file format. These files contained the main geometry, material, loading and evaluation settings for the model variant. These settings were used to generate a consistent COMSOL model definition, including the required geometry operations, material assignments, selections, study settings and post-processing outputs. An overview of the workflow is shown in Figure 1.

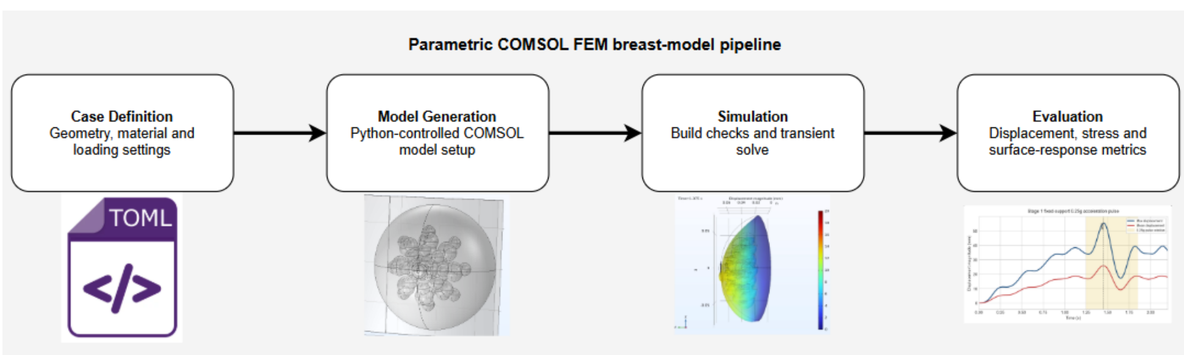


Figure 1: Overview of the COMSOL-based model-development workflow. TOML case definitions specify the main geometry, material and loading settings. These settings are used by the Python workflow to generate the COMSOL model, perform build and solve steps, and export compact evaluation outputs for comparison.

The purpose of this workflow was to make model changes traceable. Instead of manually modifying a COMSOL model for each comparison, the relevant modelling assumptions were stored in version-controlled case definitions. This allowed related model variants to be generated from a consistent structure and made it easier to identify which modelling change was responsible for a difference in the simulated displacement or stress response. The project repository contains the implementation of the Python workflow, the active TOML case definitions and the compact evaluation outputs used for the comparisons [25]. These case definitions were used to keep the modelling assumptions for each variant traceable without relying on manual edits to a single COMSOL model file.

3.1.3 Model-development strategy

The model was developed by adding and testing one modelling component at a time. The development was organised around six main modelling themes: dynamic excitation, posterior chestwall geometry, internal fibroglandular tissue distribution, outer-shape variation, mechanical support representation and tumor-overlay implementation. This structure was used to reduce the risk of interpreting multiple model changes at once.

The first part of the workflow focused on establishing a stable dynamic input and an anatomical reference geometry. The posterior support was then refined using a curved chestwall representation, followed by a more detailed fibroglandular tissue distribution. After this anatomical route was established, additional model variants were used to investigate outer-shape asymmetry, Cooper-ligament support approximations, and skin and material sensitivity. The tumor-overlay implementation was treated as a separate final extension of the workflow. In this extension, spherical tumor-overlay definitions were prepared for different sizes and locations to test whether local stiffness inclusions could be added consistently to the existing COMSOL model. Figure 2 summarises this development structure.

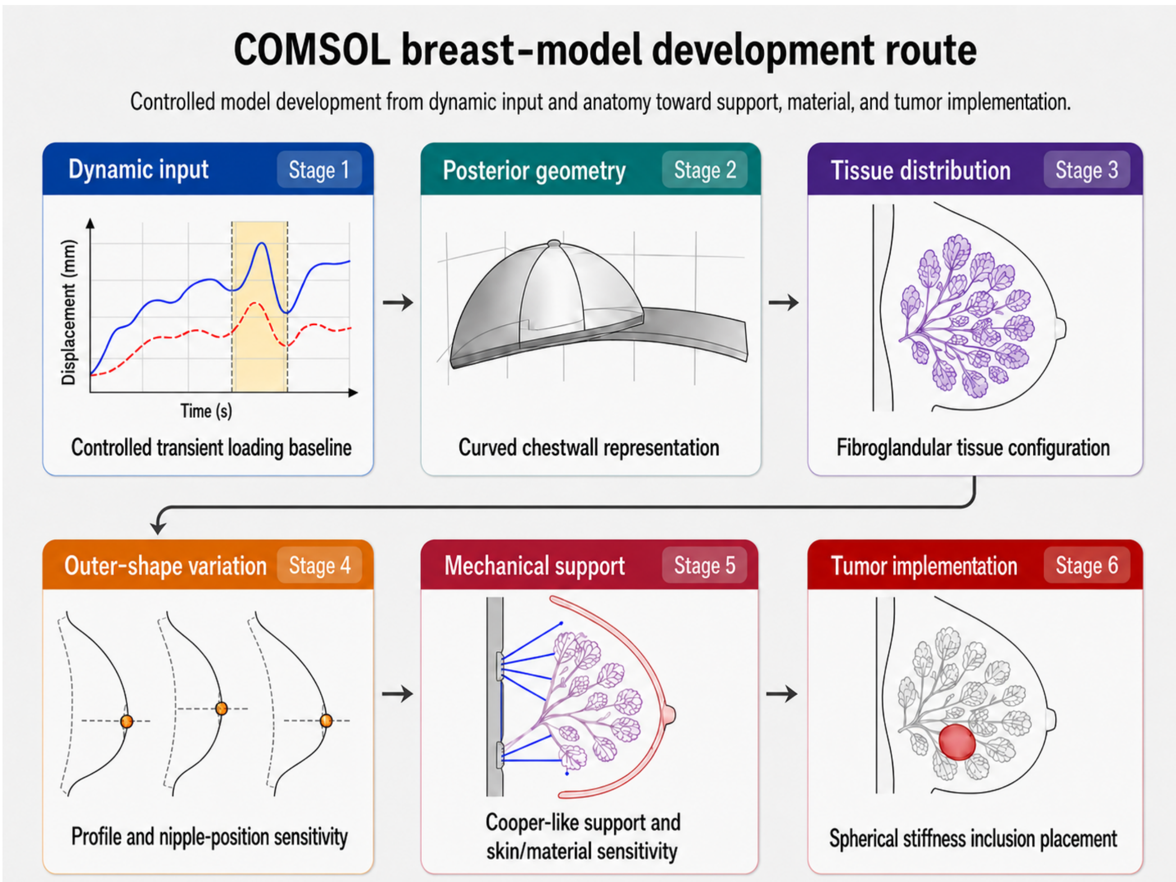


Figure 2: Schematic overview of the COMSOL breast-model development route used in this project. The workflow progresses from controlled loading inputs and anatomical model definition toward outer-shape variation, Cooper-like support, skin and material sensitivity, and tumor-overlay implementation. The blocks represent model-development themes rather than equally complete quantitative result sets; some variants were used as solved comparison cases, while others were retained as scout or build-level implementation checks.

3.2 Anatomical model definition

The anatomical model was defined as a parametrised breast representation. This made it possible to vary breast geometry, tissue distribution and local inclusions in a defined way while keeping the model construction reproducible. Patient-specific geometries remain important for clinical breast modelling [16, 17, 19], but the present work focused on a reusable model-development platform.

In this section, the anatomical definition is limited to the geometric and tissue-layout components of the model. This includes the parametrised breast envelope, posterior chestwall representation, fibroglandular tissue distribution, and defined outer-shape and nipple-position variations. Anatomically motivated components that were implemented mainly through material or support definitions, such as the skin layer, Cooper-ligament support approximation and tumor inclusion, are described in Section 3.3.

3.2.1 Parametrised breast geometry and reference volume

The overall breast shape was generated from the TOML case-definition parameters used by the COMSOL pipeline. For the selected anatomical route, the geometry used a breast radius of 70 mm, an elliptic outer profile, an anterior projection scale of 1.08 and a superior-pole scale of 0.98. In this parametrisation, the breast radius controls the overall breast size, the elliptic profile defines the basic outer contour, the anterior projection scale adjusts the forward extent of the breast, and the superior-pole scale adjusts the upper-breast fullness. These values were used as reference model-construction settings because they produced a smooth breast envelope with a plausible natural shape and a volume within the selected reference-volume range.

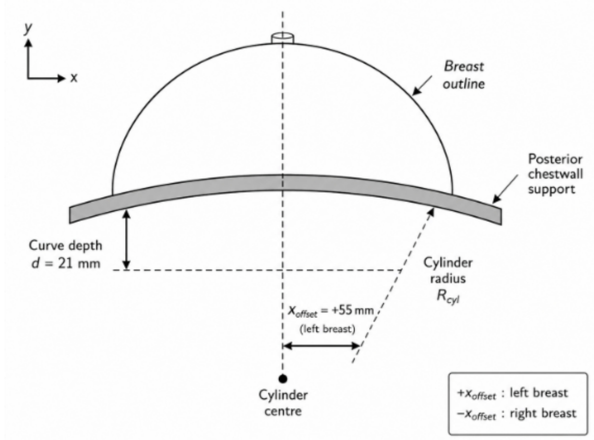
The reference volume was chosen by comparing the generated COMSOL geometry with literature values for breast-volume measurement and with an internal Catharina Hospital dataset of 100 manually delineated breast volumes [26–29]. The details of this comparison are provided in Appendix C. Because breast-volume estimates depend on measurement method, patient position and boundary definition, the selected volume was not treated as a single population average, but as a defensible reference size for the model.

After the COMSOL geometry operations, including posterior support construction and volume-preserving compensation, the selected geometry produced a breast volume of $\sim$585 mL. This value is within the lower-mid range of the internal dataset and is consistent with the literature-based volume context. It was therefore used as the reference volume for the main matched comparisons. Keeping this volume fixed prevented later changes in tissue distribution, asymmetry, support representation or tumor inclusion from being combined with a change in overall breast size.

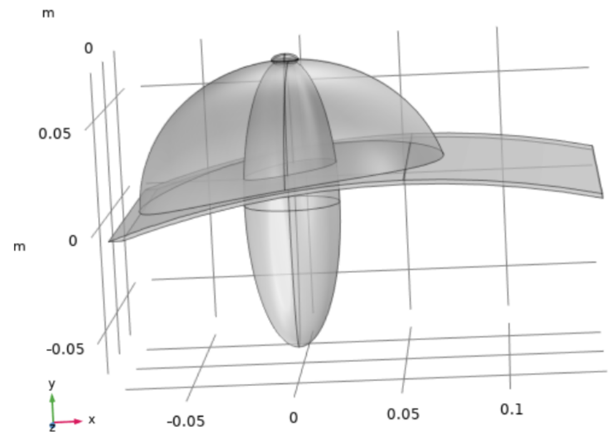
3.2.2 Posterior chestwall representation

The posterior boundary of the breast was represented by a simplified chestwall support geometry. This support was not intended as a full anatomical reconstruction of the thoracic wall or pectoral region, but as a controlled posterior interface that better reflects the curved support on which the breast rests than the flat chestwall used in earlier internal EWS modelling work. Curved posterior support is also consistent with breast FEM literature, where the breast is commonly modelled relative to the thoracic wall, pectoral region or posterior support boundary [17, 19].

In the selected reference route, the posterior support was implemented as a transverse curved support geometry with volume-preserving compensation, as shown in Figure 3. The support surface was constructed as a curved cylindrical band behind the breast. Its shape was controlled by two main parameters: the curve depth and the lateral position of the cylinder centre. The curve depth defines how far the curved support deviates from a flat posterior support, while the cylinder-centre offset shifts the curvature laterally. The selected reference configuration used a curve depth of 21 mm and a cylinder-centre offset of +55 mm in the COMSOL x-direction. In this report, the positive offset corresponds to the selected left-breast orientation; the same construction can be mirrored using a negative offset for a right-breast orientation.



(a) Schematic top view



(b) COMSOL-rendered superior view

Figure 3: Posterior chestwall representation used in the selected reference route. The schematic top view illustrates how the posterior support is generated from a curved cylindrical band. The COMSOL-rendered superior view shows the corresponding curved posterior support geometry for the selected +55 mm left-breast configuration.

Superior–inferior curvature can also be combined with the transverse support geometry in the pipeline. However, this additional curvature was not retained in the selected reference baseline because the transverse curvature represented the dominant anatomical refinement compared with the earlier flat chestwall, while exploratory COMSOL checks showed only a limited additional effect from the superior–inferior component. Excluding this extra curvature also kept the reference geometry simpler and more stable for the matched comparisons. When the chestwall shape was changed, the nipple-related helper geometry and the anterior glandular alignment were updated automatically so that the internal tissue layout remained consistent with the selected breast geometry.

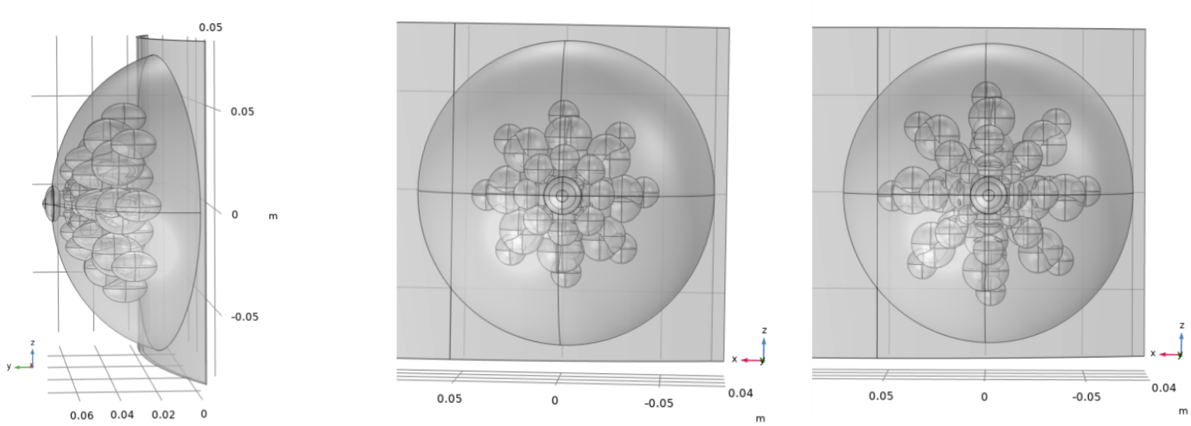
3.2.3 Fibroglandular tissue distribution

The fibroglandular tissue representation was refined from a simple internal region toward a more structured lobule-based distribution. This step was included to make the internal tissue layout more anatomically realistic, because breast tissue is heterogeneous and internal tissue composition can influence the simulated deformation response [18, 19].

The selected reference route was guided mainly by the multi-component breast-model approach described by Chen et al. [18]. In the COMSOL implementation, the fibroglandular region was generated using the `chen_2024_template_lobes` mode. The reference configuration used 12 lobules, divided into 4 inner and 8 outer lobules, with duct-like components directed toward the nipple, a subareolar core and a subareolar bridge. A lateral view of the reference configuration is shown in Figure 4a.

Three spread variants were retained: compact, reference and wide. These variants changed the spatial spread of the lobules while keeping the selected Stage 2 chestwall geometry and the same general model setup. The reference spread was used as the main anatomical route for later comparisons, while the compact and wide variants were used to check the geometric effect of the assumed fibroglandular spread. The compact and wide variants are shown in Figure 4b.

The current reference model had a total breast volume of ∼585 mL and used a glandular fraction of ∼24.3%, corresponding to a glandular volume of ∼142.2 mL. This value was considered acceptable for the reference route after comparison with literature on volumetric breast density and fibroglandular tissue quantification [30–32]. Because volumetric fibroglandular fraction is not directly equivalent to projected mammographic density or to the Breast Imaging Reporting and Data System (BI-RADS) breast-composition categories, the literature context is summarised separately in Appendix D [33]. The resulting reference and spread-variation layouts are shown in Figure 4.



(a) Reference lateral view

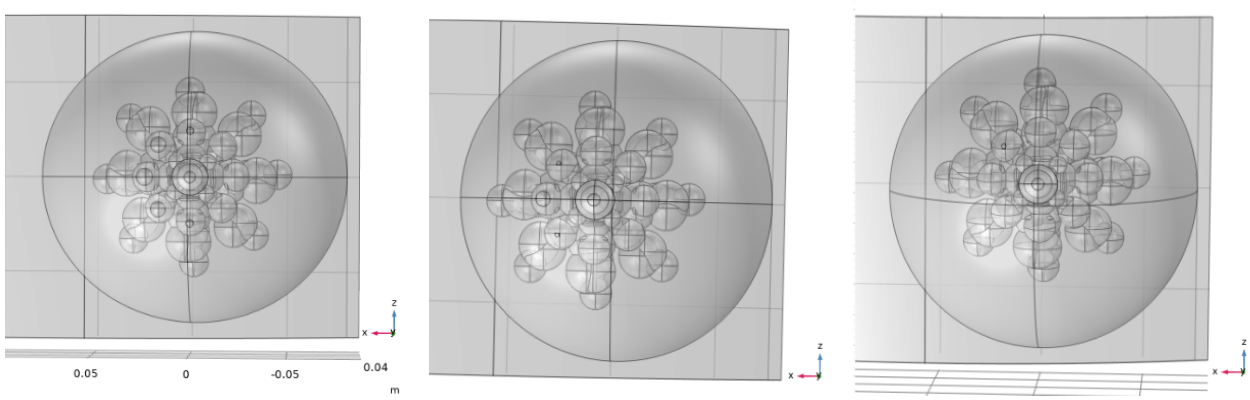
(b) Compact (left) and wide (right) anterior views

Figure 4: COMSOL-rendered fibroglandular tissue configurations used in the Stage 3 anatomical route. The reference lateral view shows the chestwall-aware lobule and duct structure inside the selected +55 mm chestwall-offset geometry. The compact and wide anterior views illustrate the spread variants used to evaluate the effect of fibroglandular spatial distribution.

3.2.4 Outer-shape and nipple-position variations

Outer-shape and nipple-position variations were included because breast size, shape and left–right symmetry vary substantially between individuals [19, 26]. For an EWS-oriented model, this is relevant because surface shape and nipple position influence both the visible breast surface and the interpretation of local displacement or landmark motion.

Stage 4 was therefore used to test geometry variations on top of the selected Stage 2 chestwall geometry and Stage 3 fibroglandular reference. In the COMSOL case definitions, the outer breast shape can be scaled in width and height, while the nipple position can be shifted over the breast surface in the x- and z-directions. Examples of these geometry variations are shown in Figure 5.



(a) Profile scaling (widening)

(b) Lateral nipple shift

(c) Superior nipple shift

Figure 5: COMSOL-rendered examples of the outer-shape and nipple-position variation cases used in Stage 4. The cases test controlled changes in breast profile and nipple position while preserving the selected Stage 2 chestwall route and Stage 3 fibroglandular reference.

3.3 Mechanical model definition

The mechanical model defines the material behaviour and support assumptions assigned to the anatomical regions described in Section 3.2. The main mechanical components were the soft-tissue material definitions, volumetric skin layer, Cooper's ligaments and tumor-overlay material modification.

3.3.1 Constitutive material model

Breast tissue was modelled as a soft, nearly incompressible material. This is consistent with common breast FEM practice, although reported mechanical properties vary substantially between studies because of differences in measurement protocol, preload, strain level, tissue type and constitutive model [4–6, 21]. In the COMSOL source-case definition, adipose, fibroglandular, skin and tumor materials were parameterised using density, bulk modulus K and two Mooney–Rivlin-type coefficients, C_{10} and C_{01}. Related Mooney–Rivlin-type soft-tissue material descriptions were also used in previous internal EWS/FEBio modelling work, which served as historical reference material for the current COMSOL route [23, 24].

The source material parameterisation can be written in terms of a deviatoric Mooney–Rivlin strain-energy density with an additional volumetric contribution [34, 35],

$$W = C_{10}\left(\bar{I}_1 - 3\right) + C_{01}\left(\bar{I}_2 - 3\right) + W_{\mathrm{vol}}\left(J; K\right). \tag{1}$$

Here, $\bar{I}_1$ and $\bar{I}_2$ are the modified invariants, J is the volume ratio and the volumetric part is controlled by the bulk modulus K. Equation 1 describes the source material parameterisation used in the workflow. It should not be interpreted as a full derivation of the exact volumetric penalty formulation used internally by COMSOL.

For stiffness-scale interpretation, and for the effective small-strain material route used where the COMSOL implementation requires linear-elastic quantities, the Mooney–Rivlin source parameters were mapped to equivalent small-strain elastic constants using:

$$G_0 = 2\left(C_{10} + C_{01}\right), \tag{2}$$

$$E = \frac{9KG_0}{3K + G_0}, \tag{3}$$

and

$$\nu = \frac{3K - 2G_0}{2\left(3K + G_0\right)}. \tag{4}$$

The coefficients C_{10} and C_{01} determine how strongly the Mooney–Rivlin material resists shape change at small deformation. Their sum gives the small-strain shear modulus G_0. This shear modulus was then combined with the bulk modulus K, which controls resistance to volume change, to obtain an equivalent Young's modulus E and Poisson's ratio ν for small-strain interpretation. Because soft breast tissue was treated as nearly incompressible, K was chosen much larger than G_0, resulting in ν close to 0.5.

For nearly incompressible behaviour, where K is much larger than G_0, this reduces approximately to

$$E \approx 3G_0 = 6\left(C_{10} + C_{01}\right). \tag{5}$$

The resulting stiffness value E was therefore reported as a small-strain equivalent stiffness, allowing the material sets to be compared more directly with each other and with literature values commonly reported in terms of Young's modulus. It does not replace the original Mooney–Rivlin source-parameter definition. The posterior chestwall support was represented separately using a linear elastic surrogate material with $E = 10$ kPa and $\nu = 0.49$.

3.3.2 Material parameter sets

The material parameter sets were defined to support both a main reference configuration and additional stiffness-sensitivity cases. The reference values were selected from the literature comparison and internal model-justification notes, with the aim of keeping the adipose and fibroglandular stiffnesses within the low-kPa range commonly used for breast soft-tissue FEM models [4–6, 21]. A summary of the literature values considered during parameter selection is provided in Appendix E.

In addition to the reference set, lower- and higher-stiffness variants were tested using the same anatomical geometry and boundary-condition route. These variants were included to evaluate the sensitivity of the simulated mechanical response to the selected material stiffness.

The main material definitions used in the COMSOL model-development workflow are summarised in Table 1. The listed Young's modulus values are included as small-strain equivalent stiffness values, so that the material scale can be compared more easily across parameter sets and with literature values reported in terms of Young's modulus.

Table 1: Main material parameters used in the COMSOL model-development workflow.

Component	Density	Bulk / elastic setting	Material coefficients	Stiffness scale
Adipose tissue	$950\ \mathrm{kg\,m^{-3}}$	$K = 425$ kPa	$C_{10} = 310$ Pa; $C_{01} = 300$ Pa	$E \approx 3.66$ kPa
Fibroglandular tissue	$1070\ \mathrm{kg\,m^{-3}}$	$K = 425$ kPa	$C_{10} = 833$ Pa; $C_{01} = 833$ Pa	$E \approx 10.0$ kPa
Skin layer	$1100\ \mathrm{kg\,m^{-3}}$	$K = 8.33$ MPa	$C_{10} = 41667$ Pa; $C_{01} = 41667$ Pa	$E \approx 500$ kPa
Hard tumor overlay case	$1079\ \mathrm{kg\,m^{-3}}$	$K = 425$ kPa	$C_{10} = 8333$ Pa; $C_{01} = 8333$ Pa	$E \approx 100$ kPa
Chestwall support	$1050\ \mathrm{kg\,m^{-3}}$	$E = 10$ kPa; $\nu = 0.49$	Linear elastic support material	$E = 10$ kPa

3.3.3 Volumetric skin representation

The skin was treated as a separate mechanical component because its stiffness can be substantially higher than that of the internal adipose and fibroglandular tissues, making it relevant for whole-breast deformation [21, 36]. In this model, the skin layer mainly represents the mechanically stiffer outer skin tissue, rather than a detailed anatomical model of the epidermis, dermis and deeper soft-tissue layers. Subcutaneous fat was therefore not defined as a separate layer and can be considered part of the adipose tissue representation.

During model development, both surface-based shell and volumetric solid-layer representations were considered. Shell-based skin descriptions were used in previous internal EWS/FEBio work [23, 24] and are also used in breast finite-element models reported in the literature [36]. A volumetric layer was retained for the COMSOL route because it allowed the skin to be defined through the same material parameter workflow as the internal breast tissues.

In the active skin-enabled cases, the volumetric skin layer was assigned a thickness of 1.5 mm. This value is consistent with reported breast skin thickness measurements and modelling assumptions in the literature, where average values around 1.45–1.55 mm and modelled skin thicknesses in the same range have been reported [17, 36–38]. Reported skin thickness and stiffness vary with anatomical location, subject characteristics and measurement method [5, 21, 37, 38]. The skin material used the density, bulk modulus and Mooney–Rivlin-type coefficients listed in Table 1.

Because reported skin stiffness values differ substantially between studies and depend on the measurement method and modelling assumptions, the skin stiffness was also varied in sensitivity cases [5, 21]. These variants were compared with the no-skin model to evaluate the mechanical influence of adding a distinct skin layer. Earlier COMSOL shell/scaffold skin attempts were not retained as the main implementation route because they produced unrealistic breast-surface behaviour in the generated models and were less straightforward to control than the volumetric solid-layer implementation.

3.3.4 Cooper-ligament support approximation

Cooper-ligament support was represented as a simplified mechanical support contribution rather than as a reconstructed anatomical ligament network. This was chosen because Cooper's ligaments are difficult to segment and assign reliable material properties, and reduced support representations are commonly used in breast finite-element modelling [19,21,36]. In the COMSOL route, the support contribution was implemented through displacement-proportional restoring tractions on selected anterior support regions.

Three support-selection variants were implemented: nipple-region support directed toward the posterior chestwall, skin-to-glandular support, and a denser skin-to-glandular support selection. These variants were used to compare simplified anterior support assumptions within the same model framework.

The support scale was defined from an assumed ligament stiffness, an active area fraction and a reference length,

$$k_{\mathrm{Cooper}} = \frac{E_{\mathrm{Cooper}} f_{\mathrm{area}}}{L_{\mathrm{ref}}}. \tag{6}$$

The active restoring traction was applied in the global anterior–posterior direction as

$$t_y = -k_y v, \tag{7}$$

where v is the COMSOL displacement component in the y-direction and k_y is the corresponding support scale. For the selection-specific support regions, the implemented scaling factors were

$$k_{\mathrm{skin}} = 0.65 k_{\mathrm{Cooper}}, \qquad k_{\mathrm{gland}} = 0.45 k_{\mathrm{Cooper}}, \qquad k_{\mathrm{nipple}} = 0.20 k_{\mathrm{Cooper}}. \tag{8}$$

The ligament stiffness scale was set to $E_{\mathrm{Cooper}} = 5.8$ MPa, based on reported tensile stiffness values for Cooper-ligament tissue summarised by Teixeira and Martins [21]. This value was not used as a direct bulk material stiffness, but was converted into an effective restoring-traction scale using Equation 6. The default support settings used $f_{\mathrm{area}} = 0.12$ and $L_{\mathrm{ref}} = 0.04$ m, giving $k_{\mathrm{Cooper}} = 17.4 \times 10^6\ \mathrm{N\,m^{-3}}$. The area fraction can be varied to define lower or higher support-strength sensitivity cases.

Because reported Cooper-ligament properties and effective whole-breast support values differ between studies, the current values should be interpreted as support-strength parameters rather than direct anatomical ligament measurements [21, 36]. For example, Chen et al. used a lower effective Cooper-ligament stiffness scale in a multi-component breast FE model, while tensile measurements of isolated ligament tissue can be in the MPa range [21, 36]. The no-Cooper model was retained as the comparison route, while the Cooper-support variants were implemented to evaluate the additional mechanical influence.

3.3.5 Tumor representation

Tumor inclusion was implemented as an analytic spherical material overlay inside the existing breast volume. The tumor was therefore not defined as a separate COMSOL geometry domain, but as a local modification of the material density and stiffness parameters inside the existing adipose and fibroglandular domains.

The tumor mask was defined as

$$m_{\mathrm{tumor}}(\mathbf{x}) = \begin{cases} 1, & \|\mathbf{x} - \mathbf{x}_{\mathrm{tumor}}\| \le r_{\mathrm{tumor}}, \\ 0, & \|\mathbf{x} - \mathbf{x}_{\mathrm{tumor}}\| > r_{\mathrm{tumor}}. \end{cases} \tag{9}$$

Here, $\mathbf{x}_{\mathrm{tumor}}$ is the tumor centre and r_{tumor} is the tumor radius. In the COMSOL implementation, this corresponds to the scalar variable `tumor_mask`, which is equal to one inside the spherical overlay and zero outside it.

The local effective material properties were then modified using the tumor mask. For a surrounding tissue with base density ρ, base Young's modulus E and base Mooney–Rivlin coefficients C_{10} and C_{01}, the effective properties were defined as

$$\rho_{\text{eff}} = \rho + m_{\text{tumor}} \left(\rho_{\text{tumor}} - \rho\right), \tag{10}$$

$$E_{\text{eff}} = E + m_{\text{tumor}} \left(E_{\text{tumor}} - E\right), \tag{11}$$

$$C_{10,\text{eff}} = C_{10} + m_{\text{tumor}} \Delta C_{10,\text{tumor}}, \tag{12}$$

and

$$C_{01,\text{eff}} = C_{01} + m_{\text{tumor}} \Delta C_{01,\text{tumor}}. \tag{13}$$

Reported lesion stiffness values span from modest benign or lower-grade pathological contrasts to higher malignant-tumor contrasts. For example, Samani et al. reported fibroadenoma and invasive ductal carcinoma values from the low-kPa to tens-of-kPa range, while elastography studies have reported malignant breast masses around or above 100 kPa [4,39,40]. Because the Early Warning Scan is aimed at detecting malignant breast tumors, a harder tumor-overlay setting was included to test a clear local stiffness contrast relative to normal breast tissue. In this high-contrast setting, the analytic overlay used $E_{\text{tumor}} = 100$ kPa together with local Mooney–Rivlin coefficient increments of $\Delta C_{10,\text{tumor}} = \Delta C_{01,\text{tumor}} = 8333$ Pa.

The tumor-overlay route was prepared for multiple tumor sizes and locations. The tested size settings used small, medium and large spherical overlays with radii of 3 mm, 6 mm and 10 mm, respectively. Build-level placements were generated for central, upper-outer, subareolar, posterior and surface-proximal positions to test whether the analytic tumor mask could be positioned consistently inside the anatomical breast model. The upper-outer placement was retained as one representative location because breast tumors are frequently reported in the upper-outer quadrant [41]. These placements defined the prepared tumor-overlay case set used for build-level implementation checks.

3.4 Dynamic loading conditions

Dynamic loading was used to excite the breast model after the mechanical geometry, material parameters and posterior support definition had been established. The loading was not intended to reproduce a complete body-motion experiment, but to provide a repeatable excitation for comparing model variants. This follows the use of simplified dynamic inputs in breast-motion modelling, where finite-element models are used to study how tissue composition, support and activity-related excitation influence breast deformation [21,36].

3.4.1 Fixed-support acceleration input

The fixed-support acceleration route used a stationary posterior support while applying an additional vertical body acceleration to the breast volume. This represents breast motion relative to a fixed torso/support frame and was used to test how the model responded to controlled acceleration amplitudes.

Before the dynamic phase, gravity was ramped to establish a preloaded gravitational deformation state. After a short hold period, the additional acceleration pulse was applied using a smooth full-sine profile,

$$a_{\text{pulse},z}(t) = \begin{cases} -A_g \, g \sin\left(2\pi \frac{t-t_0}{T}\right), & t_0 \le t \le t_0 + T, \\ 0, & \text{otherwise.} \end{cases} \tag{14}$$

Here, A_g is the acceleration amplitude expressed as a fraction of gravitational acceleration, $g = 9.81$ m s^{-2}, T is the pulse duration and t_0 is the start time of the dynamic pulse. The term $A_g \, g$ therefore gives the peak additional acceleration in m s^{-2}. The mild acceleration case used $A_g = 0.25$ and $T = 0.60$ s, corresponding to a peak additional acceleration of approximately 2.45 $\mathrm{m\,s^{-2}}$. Higher-amplitude cases were also tested to evaluate the sensitivity of the simulated response to the selected acceleration input.

A Rayleigh damping formulation was applied to the transient solid-mechanics solve to damp post-pulse oscillations and obtain a stable dynamic response for comparing model variants. The damping matrix was defined as

$$\mathbf{C} = \alpha\mathbf{M} + \beta\mathbf{K}, \qquad \beta = 0. \tag{15}$$

Here, $\mathbf{M}$ is the mass matrix, $\mathbf{K}$ is the stiffness matrix, α is the mass-proportional damping coefficient and β is the stiffness-proportional damping coefficient. The current model used $\alpha = 60\ \mathrm{s^{-1}}$ and $\beta = 0$, so only mass-proportional damping was included. This value was used as a modelling setting to obtain a damped dynamic response and was not calibrated to experimental EWS motion data.

3.4.2 Gravity preload and prescribed support displacement

The gravity-preload step was used in both dynamic routes to initialise the model from a loaded mechanical state. This is relevant because the breast response depends on the gravitational equilibrium configuration, especially for soft-tissue models with low stiffness and nearly incompressible material behaviour [19, 21].

In addition to the fixed-support acceleration route, a prescribed support-displacement route was implemented as a separate motion variant. In this route, the posterior attachment boundary was assigned a smooth vertical displacement. This is easier to relate to a person moving upward and downward, for example during a jump-like torso motion, because the support boundary itself is displaced rather than only applying an acceleration field.

The prescribed displacement was defined using a smooth bump function,

$$z_{\mathrm{support}}(t) = \begin{cases} A\,64s^3\,(1-s)^3\,, & 0 \le s \le 1, \\ 0, & \text{otherwise}, \end{cases} \qquad s = \frac{t-t_0}{T}. \tag{16}$$

Here, A is the prescribed displacement amplitude of the support and T is the duration of the motion. This bump profile was selected as a smooth model input function: it starts and ends at zero displacement with a smooth transition, while reaching its maximum at the midpoint of the motion. The factor 64 normalises the function so that its maximum equals A, because $s^3(1-s)^3$ reaches 1/64 at $s = 0.5$.

Various displacement-support variants with amplitudes of 20 mm, 40 mm and 60 mm over 0.60 s were prepared to test how explicit posterior support motion affected breast-surface response. For these cases, both absolute and support-relative displacement measures were considered. The relative vertical displacement of the support was defined as

$$w_{\mathrm{rel}}(t) = w(t) - z_{\mathrm{support}}(t), \tag{17}$$

where $w(t)$ is the vertical displacement of the evaluated breast surface. This measure separates deformation of the breast relative to the moving support from the imposed support motion itself.

3.5 Mesh, build and solver strategy

The COMSOL models were generated from scripted case definitions. Each simulation case was stored as a TOML file and translated into a COMSOL Java builder script, so that geometry settings, material parameters, loading conditions and solver settings remained reproducible between model variants.

Meshing was performed in COMSOL using a physics-controlled free tetrahedral mesh. The main reported cases used a target mesh length of 5 mm with second-order solid-mechanics elements, meaning that quadratic displacement interpolation was used within each tetrahedral element. First-order tetrahedral elements, which use linear displacement interpolation, were also available in the workflow and were useful for fast build or solver checks. However, second-order elements were retained for the reported cases because they can represent curved deformation fields more accurately within each element. The same mesh settings were used across comparable cases, so that observed differences were not intentionally introduced by changing the mesh. A full mesh-convergence study was outside the scope of this project. An example of the generated mesh is shown in Figure 6.

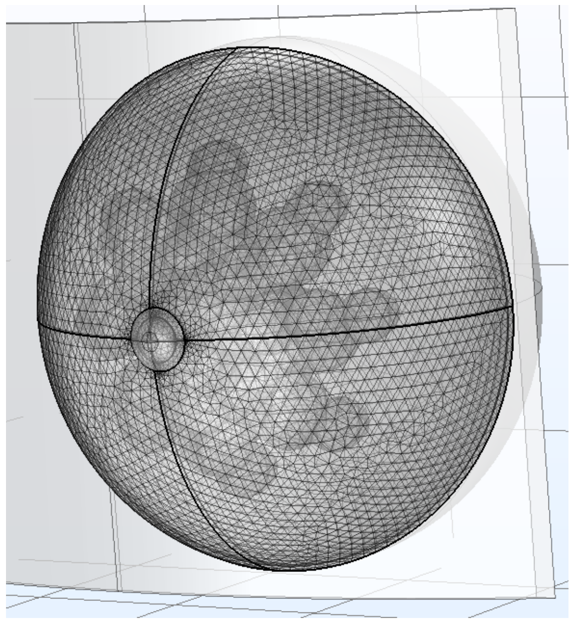


Figure 6: Example of the generated COMSOL tetrahedral mesh used for the breast model.

The solver strategy consisted of a gravity-preload step followed by the time-dependent dynamic simulation. Build-only cases were used during development to verify generated selections and model setup before running full dynamic simulations.

3.6 Post-processing and evaluation metrics

Post-processing was performed using the scripted COMSOL export and Python analysis workflow. For each completed simulation, compact output files were generated for displacement fields, selected stress quantities and time-dependent response measures. These outputs were used to evaluate how changes in model settings or parameter choices affected the simulated mechanical response, using the corresponding comparison case for each analysis.

Because the Early Warning Scan concept is based on observable breast-surface motion, the main evaluation focused on the free outer breast surface. In the COMSOL workflow this surface was represented by the `outer_skin_free_bnd` selection. Whole volume displacement and stress measures were retained as supporting diagnostic quantities.

The displacement magnitude was defined as

$$u_{\mathrm{mag}} = \sqrt{u^2 + v^2 + w^2}, \tag{18}$$

where u, v and w are the displacement components in the x-, y- and z-directions. In addition to the magnitude, signed displacement components were evaluated because they preserve direction. The vertical displacement component w was used to describe upward or downward surface motion during dynamic excitation.

For surface-based evaluation, spatial averages were computed over the evaluated surface selection S,

$$\bar{q}_S(t) = \frac{1}{A_S} \int_S q(\mathbf{x}, t)\, \mathrm{d}A, \tag{19}$$

where q is the evaluated displacement or stress quantity and A_S is the surface area of the selection. For dynamic cases, signed vertical displacement was also evaluated relative to the dynamic-start state,

$$\Delta \bar{w}_S(t) = \bar{w}_S(t) - \bar{w}_S(t_{\mathrm{dyn,start}}). \tag{20}$$

For prescribed support-displacement cases, the support-relative vertical displacement defined in Equation 17 was used in addition to the absolute displacement. Landmark-patch displacements were also exported for selected surface regions, including the nipple and superior, inferior, left and right breast-surface patches. These patch-averaged values were used as robust surface-motion measures, because they are less sensitive to individual mesh nodes than single-point displacement values.

Stress was evaluated using the exported von Mises stress field. This was used as a mechanical diagnostic for comparing stress concentration between model variants. For the tumor cases, local displacement and stress measures were additionally evaluated using the tumor mask region.

COMSOL animations were also generated for qualitative inspection of the time-dependent deformation patterns. These animations were not used as quantitative evidence; the reported Results are based on the exported displacement, stress, surface and tumor-mask metrics. The post-processing quantities used for the main model comparisons are summarised in Table 2.

Table 2: Post-processing quantities used to compare COMSOL model variants.

Quantity	Definition or source	Use in evaluation
Free-surface displacement	u, v, w and u_{mag} on `outer_skin_free_bnd`	Main EWS-relevant surface-motion response.
Surface-average response	$\bar{q}_S(t)$ and $\Delta \bar{w}_S(t)$	Average dynamic response of the visible breast surface.
Landmark-patch displacement	Patch-averaged displacement on nipple and breast-surface landmark selections	Robust comparison of visible local surface motion.
Support-relative displacement	$w_{\mathrm{rel}} = w - z_{\mathrm{support}}$	Deformation relative to prescribed support motion.
von Mises stress	Exported COMSOL stress field	Mechanical stress-concentration diagnostic.
Tumor-region response	Mask-based displacement and stress measures	Local response around the analytic tumor overlay.
Peak response	Maximum value over the evaluated time range	Quantitative comparison between corresponding cases.

When variants were compared, changes were reported relative to the corresponding comparison case,

$$\Delta q_{\%} = 100 \frac{q_{\mathrm{variant}} - q_{\mathrm{comparison}}}{q_{\mathrm{comparison}}}, \tag{21}$$

where q denotes the evaluated displacement or stress metric. This percentage difference was used only when the comparison value was non-zero and physically meaningful.

3.7 Reproducibility and data management

The project was organised to keep the COMSOL workflow reproducible and traceable. Source code, TOML case definitions, report notes and compact post-processing outputs were maintained in the project repository

[25]. Large generated files, such as full COMSOL model files, temporary solver files and extensive raw exports, were kept outside version control because they can be regenerated from the scripted case definitions.

The case-definition approach described in Section 3.1.2 was used throughout the project to keep geometry, material, loading, mesh and output settings linked to each reported model variant. This made the generated COMSOL models easier to inspect, rerun and compare.

The simulations were run using a licensed COMSOL Multiphysics installation available through TU/e. All model builds and time-dependent solves reported in this work were executed on local hardware during model development. The software and hardware environment used for the simulations is summarised in Table 3.

Table 3: Software and hardware environment used for COMSOL model development and simulation runs.

Item	Setting
COMSOL version	COMSOL Multiphysics 6.4
License environment	TU/e licensed COMSOL installation
Operating system	Windows 11 Enterprise
Processor	11th Gen Intel Core i7-11800H, 2.30GHz
Memory	16.0 GB RAM
Python environment	Python 3.8.8
Repository	[25]

4 Results

4.1 Overview of completed model variants

The Results chapter focuses on the model variants that were generated to evaluate the developed COMSOL breast model. Table 4 summarises which result groups were available and how they are used in the following subsections. This distinction is included because some routes produced complete quantitative outputs, while others were retained as scout, diagnostic or build-level outputs. In this context, scout and diagnostic outputs were used to check model behaviour or implementation, while build-level outputs only confirmed generated geometry or placement without a complete dynamic comparison.

Table 4: Overview of model-variant groups used in the Results chapter.

Result group	Available output	Use in this Results chapter
Reference anatomical model	Solved anatomical route with selected chestwall geometry and realistic glandular distribution	Establishes the main anatomical model used for later comparisons
Material and skin variants	Quantitative displacement and stress outputs for selected material and volumetric-skin cases	Used to compare material and skin-related response changes
Cooper-like support	No-Cooper comparison case and Cooper-like support scout outputs	Used to report current support-sensitivity status
Dynamic loading	Solved fixed-support acceleration cases and prescribed support-displacement output	Used to compare loading-route response measures
Tumor overlay	Build-level placement examples, simple-gland tumor metrics and lightweight realistic-route tumor output	Used to report tumor-mask implementation and available local tumor-response metrics

The following subsections report quantitative trends only when the output status supports such comparison. Diagnostic, build-level or incomplete cases are reported as implementation status rather than as final quantitative trends.

4.2 Reference model response

The developed COMSOL workflow produced a solvable anatomical breast model with the selected curved chestwall geometry and realistic glandular distribution. This model was used as the anatomical basis for the sensitivity comparisons in the following subsections. The reference route retained a breast volume close to 585 mL and increased the glandular fraction to approximately 24.3% after introducing the realistic glandular tissue distribution.

The generated analysis outputs included displacement fields, signed vertical surface response, displacement statistics and von Mises stress fields. These outputs were used to check that the model produced a measurable dynamic response under the fixed-support acceleration input before individual material, skin, Cooper-like support, loading and tumor-overlay variants were compared.

Time-dependent displacement and stress visualisations were also generated during post-processing. These outputs were used for visual inspection of the deformation pattern and local stress distribution, while the quantitative comparisons in the following subsections are based on the exported displacement and stress metrics.

4.3 Effect of material-parameter variation

Material-parameter variation was evaluated using skin-enabled full-glandular model variants with different internal tissue and skin stiffness settings. The peak displacement and von Mises stress values are summarised in Table 5.

Table 5: Material-parameter variation results for skin-enabled model variants. VM denotes von Mises stress.

Case	Peak avg. disp. (mm)	Peak max. disp. (mm)	Peak avg. VM (kPa)	Peak max. VM (kPa)
Volumetric skin, 1.25g	3.4	5.9	1.4	51.6
Volumetric skin + soft interior, 1.25g	5.9	10.9	1.8	72.4
Volumetric skin 88 kPa + soft interior, 1.25g	13.7	24.8	1.3	44.3
0.1 mm soft skin + soft interior, 1.25g	27.2	60.0	1.0	44.6

Within the skin-enabled variants, the peak maximum displacement increased from 5.9 mm for the volumetric-skin case to 10.9 mm when the internal tissue stiffness was reduced. The lower-stiffness skin variants produced larger peak maximum displacements of 24.8 mm and 60.0 mm. Peak maximum von Mises stress ranged from 44.3 kPa to 72.4 kPa across the same material-parameter set.

A separate simple-gland skin/material post-processing set was used to compare the relative influence of skin thickness and stiffness with matched `ews_surface` outputs. Relative to the 1.5 mm soft volumetric-skin reference, the tested variants changed peak displacement by approximately +35.0% to −75.6% (Figure 7).

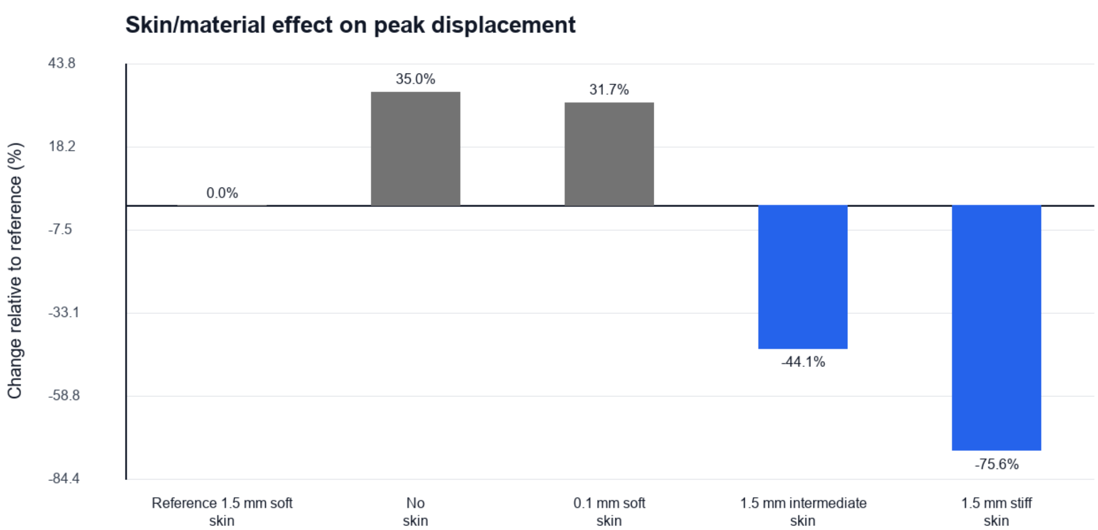


Figure 7: Relative change in peak breast displacement for the simple-gland skin/material sensitivity cases, reported with respect to the 1.5 mm soft volumetric-skin reference.

4.4 Effect of volumetric skin representation

The volumetric skin representation was evaluated on the same full-glandular model route by comparing the no-skin case with skin-enabled variants at the same 1.25g dynamic input. The skin-enabled cases used a 1.5 mm volumetric skin layer. The peak displacement and von Mises stress values are summarised in Table 6.

Table 6: Comparison of no-skin and 1.5 mm volumetric-skin model variants at 1.25g. VM denotes von Mises stress.

Case	Peak avg. disp. (mm)	Peak max. disp. (mm)	Peak avg. VM (kPa)	Peak max. VM (kPa)
No skin, 1.25g	7.3	14.2	0.8	3.9
Volumetric skin, 1.25g	3.4	5.9	1.4	51.6
Volumetric skin + soft interior, 1.25g	5.9	10.9	1.8	72.4

Compared with the no-skin case, the volumetric-skin case reduced peak average displacement from 7.3 mm to 3.4 mm and peak maximum displacement from 14.2 mm to 5.9 mm. With the soft-interior material variant, peak average displacement increased to 5.9 mm and peak maximum displacement increased to 10.9 mm. Peak maximum von Mises stress was higher in both skin-enabled variants than in the no-skin case. These higher stress values are interpreted as model stress-concentration metrics, especially near stiff material interfaces and the posterior support region.

An example von Mises stress field for the Stage 5 no-Cooper 1.25g case is shown in Figure 8. The highest stress values in this example were located near the posterior chestwall/support attachment region.

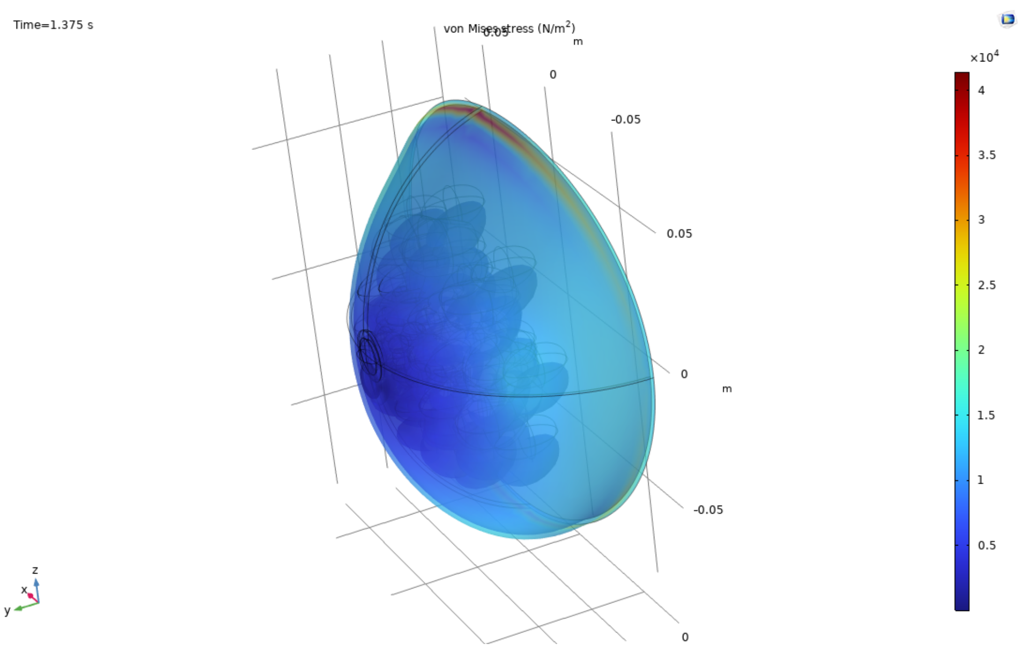


Figure 8: Example von Mises stress distribution for the Stage 5 no-Cooper 1.25g case at $t = 1.375$ s. The highest stress values are located near the posterior chestwall/support attachment region, while the free breast volume shows lower stress values.

4.5 Effect of Cooper-like support

The Cooper-like support approximation was first evaluated against the no-Cooper comparison case. In the anatomical Stage 5 route, the no-Cooper case reached the expected final time of 2.2 s, while the default Cooper-support case failed immediately during the dynamic solve because of solver-capacity limitations. The failed case was therefore not used for quantitative response comparison.

A separate simple-gland Cooper-support scout was used to compare the implemented support-selection variants. The resulting peak displacement values are summarised in Table 7.

Table 7: Cooper-like support scout results for the simple-gland model route.

Case	Peak avg. disp. (mm)	Peak max. disp. (mm)	Avg. disp. change (%)	Max. disp. change (%)
No Cooper baseline	22.55	45.58	0.00	0.00
Gland-to-skin area 0.03	22.44	44.99	-0.50	-1.29
Gland-to-skin area 0.06	22.44	44.97	-0.52	-1.34
Gland-to-skin area 0.12	22.43	44.96	-0.53	-1.37
Nipple-to-chest area 0.03	22.55	45.64	-0.03	0.12
Dense network area 0.03	22.44	44.99	-0.50	-1.29

In the simple-gland scout, the gland-to-skin support variants changed peak average displacement by -0.50% to -0.53% and peak maximum displacement by -1.29% to -1.37% relative to the no-Cooper baseline.

4.6 Effect of dynamic loading route

4.6.1 Fixed-support acceleration amplitude

The fixed-support acceleration response was evaluated on the no-Cooper, +55 mm chestwall-offset, full-glandular model route. The $0.25g$ and $0.50g$ cases reached the expected final time of 2.2 s, while the $0.75g$ case did not produce a valid solved output. The compact review metrics are summarised in Table 8.

Table 8: Dynamic-amplitude results for the fixed-support acceleration route.

Case	End time (s)	Review avg. disp. (mm)	Surface signed w (mm)	Output status
No Cooper, +55 mm offset, $0.25g$	2.2	3.0	0.16	Solved
No Cooper, +55 mm offset, $0.50g$	2.2	2.9	0.31	Solved
No Cooper, +55 mm offset, $0.75g$	–	–	–	Failed/not valid

The surface signed vertical response increased from 0.16 mm for the $0.25g$ case to 0.31 mm for the $0.50g$ case.

4.6.2 Prescribed support-displacement route

A prescribed support-displacement route was evaluated as a separate motion variant. This case used the +55 mm chestwall-offset geometry with a simple glandular structure, no Cooper-like support, a 1.5 mm volumetric skin layer and soft internal material settings. For the 40 mm, 0.60 s support-displacement case, support-relative displacement metrics were extracted from the free outer surface and nipple landmark region. The extracted support-relative metrics are summarised in Table 9.

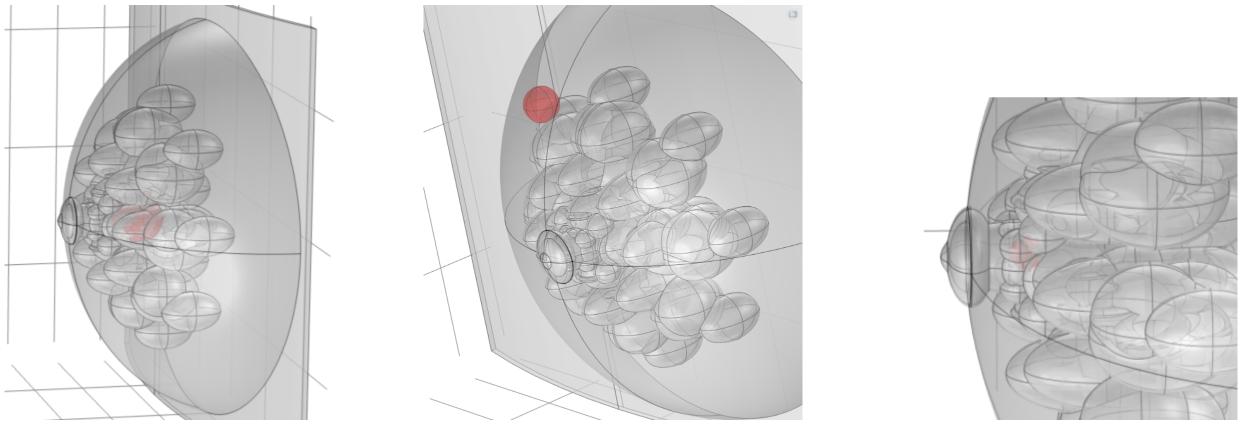
Table 9: Prescribed support-displacement response for the 40 mm, 0.60 s motion variant.

Metric	Maximum value	Selection
Support-relative spatial average of w	8.81 mm	Free outer surface
Support-relative maximum of w	15.70 mm	Free outer surface
Support-relative spatial average of w	15.51 mm	Nipple landmark region

This route was evaluated as a separate motion variant and was not directly compared with the fixed-support acceleration cases, because the imposed motion definition and model basis were different.

4.7 Tumor-overlay placement

The tumor-overlay route was implemented as a build-level model variant. The analytic tumor mask could be placed inside the existing breast volume and visualised for different tumor positions and sizes. Figure 9 shows representative central, upper-outer surface-proximal and subareolar tumor-overlay placements.



(a) Large central (b) Medium upper-outer surface-proximal (c) Small subareolar

Figure 9: Representative tumor-overlay placements generated in COMSOL. The breast volume and glandular structures are shown transparently, while the analytic spherical tumor overlay is highlighted in red.

The build-level tumor variants included central, upper-outer, subareolar, posterior and surface-proximal placements. These variants confirmed that the tumor mask could be positioned and visually inspected within the anatomical breast model. Complete dynamic tumor-overlay sensitivity results were not retained as final quantitative result cases in this report. The tumor-overlay results are therefore limited to implementation and placement checks; quantitative tumor-sensitivity analysis requires a larger solved case set with matched no-tumor controls and complete tumor-mask post-processing.

4.8 Summary of main result trends

The strongest quantitative sensitivity was observed for the material and skin-parameter variants. Across the simple-gland skin/material result set, peak displacement changed by approximately +35.0% to −75.6% relative to the selected reference case. The volumetric skin layer and its stiffness therefore had a larger effect on the global displacement response than the other tested model routes.

The simplified Cooper-like support variants produced a much smaller global response change. In the available scout results, global peak displacement changed by approximately −1.4% to +0.1% relative to the no-Cooper baseline. This indicates that the current Cooper-like support implementation mainly served as a controlled support-sensitivity check rather than as a dominant mechanical effect.

The dynamic-loading results showed that both the fixed-support acceleration route and the prescribed support-displacement route could be evaluated, but these routes should be interpreted separately because their imposed motion definitions are different. The tumor-overlay route was implemented at build level for several tumor sizes and locations, but complete dynamic tumor-sensitivity results were not retained as final quantitative results. Additional time-dependent comparison plots for the skin/material, model-development and dynamic-loading outputs are provided in Appendix F.

5 Discussion

5.1 Contribution to an EWS-oriented FEM modelling workflow

This project developed a progressively more complete COMSOL finite-element route for simulating dynamic breast deformation in the context of the Early Warning Scan (EWS). The main contribution was not a clinically validated detection model, but a reproducible model-development framework in which anatomical structure, material parameters, support assumptions, loading route and tumor-overlay placement can be varied in a controlled way. This distinction is important because EWS is based on observable breast-surface motion, whereas many of the model quantities used during development, such as internal stress or volume-averaged displacement, are primarily mechanical diagnostics.

Compared with the earlier simplified representation, the final route made the model more anatomically and mechanically configurable. This made it possible to evaluate which components most strongly affected the simulated surface response, while still treating the present results as model-development outcomes rather than evidence of diagnostic performance.

5.2 Anatomical realism and glandular representation

A central aim of this project was to improve the anatomical realism of the FEM breast model. The realistic glandular route added a spatially distributed fibroglandular structure instead of representing the internal breast tissue with a single simplified region. This is relevant because breast deformation depends not only on total breast size, but also on the relative amount and spatial distribution of adipose and fibroglandular tissue [4–6, 21].

The current glandular implementation also introduced a practical limitation. A detailed glandular structure is mechanically more representative, but it increases geometric complexity, meshing difficulty and solver cost. Similar anatomical complexity was a limiting factor in previous internal FEBio-based model-development routes [23, 24]. The current COMSOL workflow was able to build and solve the realistic glandular route, but running many dynamic variants on a local laptop remained computationally demanding. This suggests that the modelling strategy is technically feasible, but that further studies require access to stronger computational resources.

Anatomical variability should therefore become a larger part of future model development. Breast size and shape, glandular fraction, glandular asymmetry, posterior attachment geometry and tissue distribution vary strongly between individuals [5, 6, 21]. The current model can be used as a first step for building such population-level model sets, but this would require many additional variants and validation against measured breast geometry, tissue-distribution data and EWS-like surface-motion measurements. Future work should therefore vary the relevant anatomical, mechanical and loading parameters in a systematic way to quantify their influence on the simulated surface response. For future AI-based analysis, this type of systematic variation could support the construction of synthetic datasets with controlled differences in anatomy, tissue properties, loading conditions and tumor characteristics.

5.3 Material-parameter uncertainty and response realism

Material parameters remain one of the largest sources of uncertainty in breast FEM modelling. Reported stiffness values for adipose tissue, fibroglandular tissue, skin and tumor tissue vary substantially across studies because they depend strongly on experimental protocol, tissue condition and constitutive assumptions [4–6, 21]. The material settings used in this project should therefore be interpreted as controlled reference and sensitivity values rather than as patient-specific tissue properties. This is consistent with the Results, where the material and skin-parameter variants produced the largest peak-displacement changes among the reported model routes.

This uncertainty affects how the displacement results should be read. A simulated displacement amplitude is only physically meaningful when the material parameters, boundary conditions and loading route are plausible for the intended experimental situation. The current results show that the model can produce measurable dynamic surface motion, but the exact displacement scale still requires comparison with experimental or EWS-like motion data. Such validation would make it possible to determine whether the selected stiffness range reproduces realistic breast motion.

The von Mises stress results should be interpreted with caution. In this project, von Mises stress was useful as a scalar diagnostic for comparing stress concentrations between variants, especially near support and attachment regions. However, stress is not directly measured by the EWS concept and should not be treated as a direct detection output. Stress concentrations near the posterior support are more likely to reflect the imposed boundary condition and support geometry than a clinically meaningful tissue response. For EWS interpretation, surface displacement and landmark motion are therefore more relevant than internal stress metrics.

5.4 Dynamic loading and surface-motion interpretation

The dynamic loading definition is an important part of the model, because the simulated breast response depends strongly on how motion is introduced. The fixed-support acceleration input and the prescribed support-displacement route represent different idealisations of how breast motion could be generated during an EWS-like measurement. The fixed-support acceleration route applies a controlled global excitation while keeping the support geometry fixed. The prescribed support-displacement route is more directly related to a physical situation in which the chestwall or support moves up and down, for example during a jump-like motion.

In the current project, most quantitative dynamic comparisons focused on the fixed-support acceleration route. This provided a controlled way to compare model variants, but it does not necessarily mean that this route is the most realistic representation of the final EWS setup. Further work should therefore compare the fixed-support acceleration route with prescribed support-motion approaches in more detail. The resulting motion depends on the amplitude, timing, damping, gravity preload and support coupling of the model. These factors should therefore be selected or calibrated against the intended EWS motion setting, because they directly influence whether the simulated breast response is physically representative.

The current dynamic results should therefore be interpreted as development outputs rather than as a fully realistic motion experiment. The displacement time series and surface-response metrics show that the model can generate measurable dynamic motion, but the response curves still depend on the simplified excitation, damping, support assumptions and solver settings. The applied damping was used as a modelling setting rather than as a calibrated tissue-viscoelastic description, while breast soft-tissue behaviour is known to depend on measurement conditions and constitutive assumptions [5, 6]. Future work should record the motion applied in an EWS-like setup and use that measured motion, together with calibrated viscoelastic or visco-hyperelastic tissue behaviour, as input or calibration target for the FEM model.

5.5 Skin and Cooper-like support modelling choices

The skin representation had a clear mechanical role in the model because it added a relatively stiff outer layer around the softer internal breast tissues. This is consistent with multi-component breast FEM studies, where the breast is represented using separate anatomical and mechanical components rather than as one homogeneous soft-tissue volume [19, 21, 36]. In the current COMSOL route, the 1.5 mm volumetric skin layer was therefore an important step toward a more anatomically representative model. This thickness is also within reported breast-skin thickness ranges, although skin thickness and mechanical properties vary between subjects and measurement locations [37, 38].

The volumetric skin route should still be interpreted as a controlled modelling choice rather than as a patient-specific skin reconstruction. Its mechanical effect depends on skin stiffness, skin thickness and the assumed interface with the underlying adipose and fibroglandular tissue. Future work should therefore include a more systematic skin-parameter study to determine how robust the simulated surface response is across realistic skin properties.

The implementation of Cooper-like support should also be interpreted as an approximation, because the reported properties of the Cooper-ligament are limited and not directly transferable to an effective representation of breast support [21, 36]. Cooper's ligaments form a distributed fibrous support structure in the breast, while the current COMSOL route represented their influence through simplified support terms rather than explicit anatomical ligament reconstruction [36]. In the available scout results, the simplified support route produced only small global peak-displacement changes, suggesting that its current implementation mainly served as a controllable support-sensitivity test. Future versions should investigate whether a more anatomically guided support representation produces a stronger or more localised surface-response effect.

5.6 Tumor-overlay route and future tumor sensitivity

The tumor-overlay route successfully added a build-level mechanism for placing analytic spherical stiffness inclusions inside the existing breast volume. This is a useful first step because tumor size and location can be changed without rebuilding the full breast geometry for every variant. The current implementation is therefore suitable for preparing systematic tumor-sensitivity studies.

The current tumor representation is still simplified. Real breast tumors can differ in morphology, stiffness, depth and interaction with surrounding tissue, and reported tumor stiffness values vary strongly between measurement methods and tumor types [4, 7, 8, 42]. A spherical homogeneous overlay can test the effect of a controlled stiffness contrast, but it cannot represent the full range of malignant tumor morphology. This limitation is especially important for EWS, because a tumor would only be detectable through its effect on the externally observable surface motion. The relevant question is therefore not only whether a stiff inclusion changes the internal mechanical field, but whether it changes the surface response enough to be measured robustly.

The current tumor-overlay results should therefore be interpreted as implementation and placement results rather than as a completed dynamic tumor-sensitivity dataset. Future tumor studies should test a broader range of tumor geometries, stiffness contrasts and anatomical locations using complete dynamic simulations with tumor-mask post-processing, surface-motion metrics and matched no-tumor controls.

5.7 Computational limitations and reproducibility

A major limitation of the current project was that all COMSOL model builds and time-dependent simulations were run on a personal laptop. This was workable for developing the pipeline and generating geometry build-only cases, but it strongly limited the number of complete dynamic solves that could be included. For the more realistic model variants, individual dynamic simulations could take approximately 12 h per case, not including the additional time required for post-processing. This made large parameter sweeps and complete dynamic testing of the more complex model routes impractical within the project timeframe.

Access to faster external computational resources would improve the study by allowing larger sweeps, more realistic model variants and full post-processing to be run more efficiently. This would be especially important for systematic testing of material parameters, dynamic loading routes, Cooper-like support and tumor-overlay cases. The current computational limitation should therefore be interpreted as a practical limitation of the project setup and timeframe, rather than as a limitation of the modelling framework itself. Future work should account for this by running the simulations on stronger computational hardware.

The repository structure helped keep the model-development workflow traceable. The scripted case definitions, compact outputs, report notes and figure sources made it possible to track how the model variants were generated and evaluated [25]. Future work should preserve this level of traceability, although the exact repository structure can be adapted as the simulation workflow expands.

5.8 Overall perspective and future work

Overall, the project extended the COMSOL FEM breast model in several directions that are relevant for future EWS-oriented deformation studies:

- a more anatomically detailed breast geometry with chestwall curvature, asymmetry options and a realistic glandular tissue distribution;
- controlled material-parameter and volumetric-skin variants;
- simplified Cooper-like support and dynamic loading routes;
- analytic tumor-overlay placement for future stiffness-sensitivity studies;
- a traceable workflow for case definition, model building and automatic post-processing.

The most important next step is to run systematic, fully solved dynamic simulations on stronger computational hardware and compare the simulated surface response with experimental or EWS-like measurement data. This should be combined with controlled variation of material parameters, glandular anatomy, skin properties, loading route and tumor properties. If those future simulations show that tumor-related stiffness contrasts produce measurable and repeatable differences in breast-surface motion, the model could become a useful tool for exploring EWS sensitivity and design choices.

6 Conclusion

This project developed a reproducible COMSOL finite-element workflow for EWS-oriented breast deformation modelling. The final workflow supports controlled variation of anatomical geometry, tissue distribution, material parameters, skin representation, support assumptions, loading route and tumor-overlay placement.

The strongest quantitative effects were found for the material and skin-parameter variants. Volumetric skin representation and skin stiffness had a large influence on peak displacement and local von Mises stress concentrations. In contrast, the simplified Cooper-like support variants produced only small global displacement changes in the available scout results, so they should be interpreted as support-sensitivity implementation checks rather than as validated anatomical ligament models.

The tumor-overlay route was implemented at build level and visualised for different tumor sizes and locations. This confirms that analytic spherical tumor masks can be placed inside the anatomical breast model, but it does not yet provide quantitative evidence for tumor detectability. Complete tumor-sensitivity studies require matched no-tumor controls, fully solved dynamic cases, surface-based post-processing and validation against experimental or EWS-like surface-motion data. Overall, the project provides a traceable COMSOL FEM platform for future EWS model development, while showing that stronger computational resources are needed for systematic large-scale simulation and validation.

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Appendix

A Previous EWS FEM work

The present project was developed in the context of previous internal Early Warning Scan (EWS) finite element modelling work. Earlier EWS models provided a methodological starting point for parametrised breast geometry, soft-tissue material assumptions, dynamic loading and displacement-based evaluation [23, 24]. These previous projects were used as internal engineering reference material and as a comparison point for modelling choices. They were not used as validation datasets for the current COMSOL model.

The earlier FEBio-oriented modelling route used a simplified parametrised breast representation in which adipose and glandular regions were defined from geometric construction parameters. Figure 10 shows a schematic cross-section from the earlier EWS project by R. Engels [23]. This representation provided useful context for the transition toward a COMSOL-based workflow, but the current COMSOL model does not reproduce the previous geometry directly.

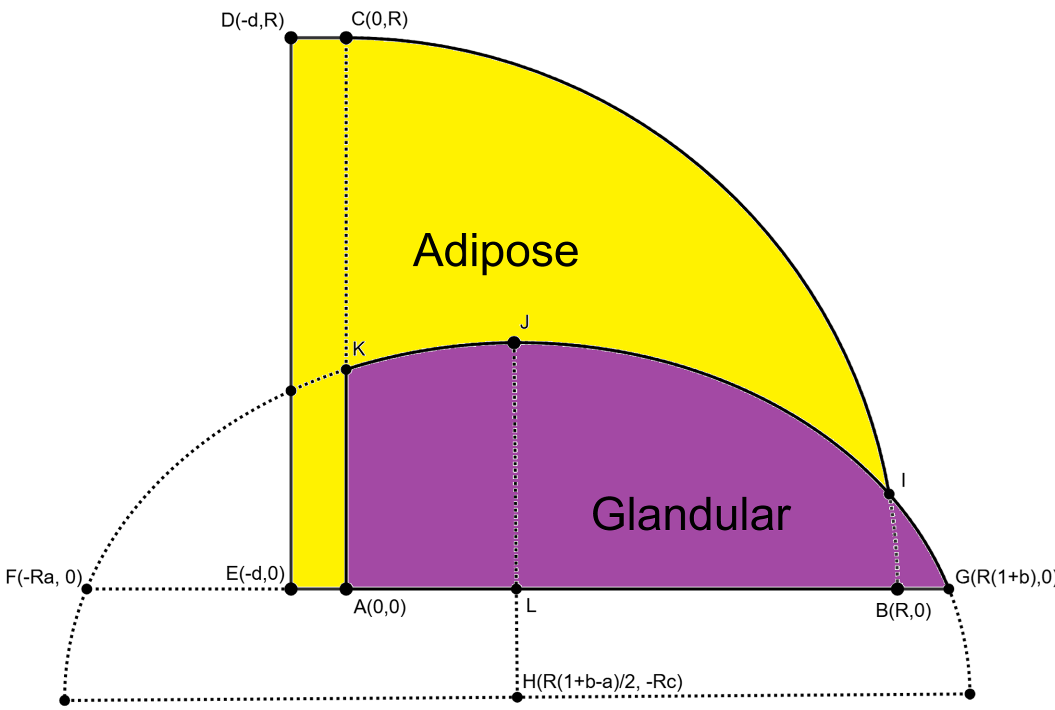

Figure 10: Schematic cross-section from the earlier EWS FEM breast model by R. Engels [23]. The figure illustrates the parametrised adipose and glandular regions used in that previous modelling route and is included here as internal methodological context.

The later EWS/SIOUX workflow provided additional context for skin representation, soft material settings and distributed reinforcement concepts for Cooper-ligament-like support [24]. In the current project, these earlier modelling choices were treated as reference points rather than as fixed requirements. The COMSOL route was developed as a separate implementation with emphasis on controlled geometry construction, automated region definitions, build-only checks and systematic comparison between model variants. This made it possible to investigate selected anatomical and mechanical modelling assumptions in a more traceable way, without implying that the current model is a validated patient-specific representation.

B Current COMSOL repository and reproducibility structure

The active implementation used in this internship project is the COMSOL source-case pipeline developed during this project. The public project repository stores the reproducible parts of the work, including the Python source code, active TOML case definitions, report notes, plotting scripts, compact post-processing tables and selected figures [25]. Large generated COMSOL files, build folders, solve folders, recovery files and raw exports are not intended to be stored in the repository because they can become several gigabytes per case.

The main repository folders used for the current COMSOL workflow are summarised in Table 10. The exact folder structure may still change during repository cleanup, but the distinction between source definitions, generated artefacts and compact evaluation output was maintained throughout the project.

Table 10: Main repository components used in the COMSOL model-development workflow.

Repository component	Role in the project
`src/ews_fem_pipeline_comsol`	Active Python package for the COMSOL source-case workflow, including case loading, model generation support, Java builder generation, run control and post-processing utilities.
`runs/comsol_runs`	Version-controlled TOML case definitions for the main model variants and sensitivity studies.
`analysis_output/comsol_pipeline`	Compact evaluation outputs used for comparison, including summary tables, selected plots and post-processing results.
`docs/report_notes/comsol_pipeline`	Project notes documenting modelling choices, run status, figure sources and interpretation decisions.
`docs/Internship_report_Daan_Kuijpers`	The final internship project report.

The COMSOL simulations were run using a licensed COMSOL Multiphysics installation available through TU/e. Most model builds and dynamic solves were executed on local hardware during model development. Because time-dependent COMSOL solves can be computationally expensive, especially for second-order elements, volumetric skin layers and larger parameter sweeps, access to stronger computational resources such as TU/e high-performance computing facilities is relevant for future large-scale studies. In this project, the repository was therefore organised to preserve the reproducible case definitions and compact outputs, while allowing heavy generated artefacts to remain outside version control.

C Breast-volume reference data

An internal Catharina Hospital breast-volume dataset was reviewed to contextualise the selected COMSOL reference volume. The dataset consisted of 100 manually delineated breast volumes. The contours were drawn approximately 5 mm inside the outer skin boundary on images acquired in a supine position. The reported values should therefore be interpreted as internal reference volumes for model scaling, rather than as exact external breast volumes.

Table 11: Summary of the internal Catharina Hospital breast-volume reference dataset. The dataset contained 100 manually delineated breast volumes, with contours drawn approximately 5 mm inside the outer skin boundary in supine images.

Metric	Breast volume (mL)
Mean	728
Median	691
Standard deviation	327
25th percentile	520
75th percentile	929
Minimum	106
Maximum	1777

The selected COMSOL reference volume of ∼585 mL lies between the 25th percentile and the median of this internal dataset. This supports its use as a lower-mid reference volume for the current model-development route. Direct comparison remains limited by the segmentation definition and patient position, because the internal contours were drawn inside the skin boundary and in a supine position.

The internal dataset was also compared with literature values for breast-volume measurement. Stahl et al. reported MRI-based breast anthropometry in a cohort of 400 women and showed a broad distribution of breast volumes [26]. Yoo et al. compared MRI-estimated breast volume with mastectomy specimen weight and supports the clinical relevance of MRI-based volume estimation, although in a cancer/mastectomy cohort [27]. Nie et al. showed that skin removal can change quantitative MRI-based breast-volume and density measurements, which is relevant because the internal dataset was delineated inside the outer skin boundary [28]. Choppin et al. reviewed breast-volume measurement methods and emphasised that measurement uncertainty depends strongly on method, patient pose and boundary definition [29]. Together, these sources support using the ∼585 mL COMSOL volume as a defensible controlled reference volume, while avoiding overinterpretation as an exact population average.

D Fibroglandular fraction reference data

The selected Stage 3 reference model was compared with literature on volumetric breast density and fibroglandular tissue quantification. This comparison was used to contextualise the chosen fibroglandular fraction, not to assign a clinical BI-RADS category to the model.

Table 12: Literature context used to interpret the selected fibroglandular fraction. The model fraction refers to three-dimensional fibroglandular volume divided by total breast volume.

Source	Reported quantitative values	Relevance for this model
Sartor et al. [30]	Volumetric density groups were defined using thresholds of $< 4.5\%$, 4.5–7.5%, 7.5–15.5% and $\geq 15.5\%$. The reported volumetric breast-density range was approximately 1.9–43.3%, with a median around 7.2%.	Provides context for automated volumetric breast-density assessment and shows that the selected model fraction lies in a relatively dense range.
Nie et al. [31]	In a 3D MRI cohort of 321 women, the mean breast volume was approximately 779 cm^3, the mean fibroglandular tissue volume was approximately 86 cm^3, and the mean percent density was approximately 12.1%.	Provides MRI-based three-dimensional fibroglandular tissue measurements for comparison with the COMSOL volume fraction.
Lu et al. [32]	MRI examples showed large inter-subject variation, with example glandular tissue fractions around 20%, 40% and 60%.	Shows that substantially higher fibroglandular fractions can occur in individual dense breasts, even though they should not be treated as population-average values.
ACR BI-RADS [33]	BI-RADS breast-composition categories are qualitative clinical categories rather than direct three-dimensional volume fractions.	Supports avoiding a direct BI-RADS category assignment for the COMSOL reference model.

This density context is relevant for the EWS application because dense fibroglandular tissue can reduce lesion conspicuity and lower mammographic sensitivity in X-ray screening [2, 3]. The selected COMSOL reference case produced a breast volume of approximately 585 mL and a glandular fraction of 24.3%, corresponding to an approximate glandular volume of 142 mL. This value is higher than typical average volumetric density values reported in screening or MRI cohorts, but remains within the broader range reported for volumetric density and dense-breast examples. It should therefore be interpreted as a controlled fibroglandular-rich reference configuration for EWS-oriented model development, not as a population-average breast-density value or a direct BI-RADS category.

E Material-parameter literature summary

The material parameters used in the COMSOL model were selected as controlled reference and sensitivity values rather than as patient-specific tissue measurements. Reported breast-tissue stiffness values vary strongly between studies because they depend on measurement protocol, tissue preparation, preload, strain level, loading direction, subject characteristics and constitutive model [4–6, 21]. The values in Table 13 therefore provide stiffness-scale context rather than directly interchangeable material constants. Young's modulus values, MRE shear-stiffness values, phantom storage moduli and hyperelastic coefficients should not be interpreted as the same mechanical quantity.

Table 13: Quantitative literature context used to select and interpret the material-parameter scales in the COMSOL model.

Source	Type of value	Reported quantitative values	Relevance for this model
Samani et al. [4]	Ex-vivo Young's modulus measurements	Normal fat: 3.25 ± 0.91 kPa; normal fibroglandular tissue: 3.24 ± 0.61 kPa; fibroadenoma: 6.41 ± 2.86 kPa; intermediate-grade IDC: 19.99 ± 4.2 kPa; high-grade IDC: 42.52 ± 12.47 kPa.	Supports low-kPa normal tissue values and a higher stiffness scale for pathological tissue.
Hsu et al. [17]	Linear-elastic breast-compression FE model and literature ranges	Reported literature ranges: adipose 0.5–25 kPa, glandular 0.08–272 kPa and skin 0.088–3000 kPa. Tested FE scenarios included fat/glandular/skin stiffness combinations from 1/1/1 kPa to 10/50/100 kPa.	Shows that published breast-FE stiffness choices span orders of magnitude, supporting sensitivity testing.
Chen et al. [18, 36]	Dynamic multi-component breast FE material definitions	Adipose tissue was represented using Mooney–Rivlin coefficients including $C_{10} = 0.31$ kPa and $C_{01} = 0.30$ kPa, corresponding to $E \approx 3.66$ kPa using the small-strain estimate. Reported Young's-modulus settings included skin 500 kPa, glandular tissue 10 kPa, pectoralis/soft tissue 10 kPa and Cooper's ligaments 100 kPa.	Closest literature match for the current multi-component dynamic FE modelling route.
Mazier and Bordas [19]	Patient-specific elastic calibration	Optimised values of $E_{\mathrm{breast}} = 0.32$ kPa and $E_{\mathrm{skin}} = 22.72$ kPa were reported for shape fitting between prone and supine configurations.	Shows that calibrated effective breast and skin stiffness can be much lower than stiff-skin FE assumptions.
Aloufi et al. [43]	MRE shear stiffness in normal breasts	Fatty tissue was reported around 0.81–0.82 kPa and fibroglandular tissue around 1.46–1.55 kPa.	Supports a higher fibroglandular than fatty stiffness trend, but MRE shear stiffness is not directly equivalent to Young's modulus.
McKnight et al. [42]	Breast MRE shear stiffness	Breast carcinoma regions ranged approximately 18–94 kPa with a mean of about 33 kPa; surrounding breast tissue ranged about 4–16 kPa with a mean of about 8 kPa.	Provides tumor-stiffness contrast context for the analytic tumor-overlay route.
Kashif et al. [14]	DIET/MRE phantom material stiffness	Silicone phantom materials covered approximately 2–570 kPa. A DIET phantom used a healthy-tissue-mimicking material around 9 kPa and a tumor-mimicking inclusion around 35 kPa, giving an almost four-fold contrast.	Relevant to EWS/DIET-style surface-motion studies, but storage modulus is not identical to static Young's modulus.
Goodbrake et al. [6]	Three-dimensional anisotropic constitutive fitting	The fitted material response was not reducible to one Young's modulus. The study showed direction-dependent and heterogeneous behaviour, especially for fibroglandular tissue.	Supports treating the current isotropic material values as simplified stiffness scales rather than complete tissue descriptions.

Table 14 summarises these values by model component. This table was used as a bridge between the detailed literature values and the selected COMSOL reference and sensitivity settings. The ranges should be read as stiffness-scale context rather than as one directly comparable dataset, because the included studies use different experimental and modelling definitions.

Table 14: Summary of literature stiffness ranges used to contextualise the COMSOL material settings.

Component	Reported stiffness context	Interpretation for this project	Overall range
Adipose tissue	Reported FE/literature range of 0.5–25 kPa [17]; ex-vivo normal fat around 3.25 kPa [4]; MRE fatty tissue around 0.81–0.82 kPa [43].	Adipose tissue is generally represented in the low-kPa range, although the exact value depends on measurement method.	0.5–25 kPa
Fibroglandular tissue	Very broad reported FE/literature range of 0.08–272 kPa [17]; ex-vivo normal fibroglandular tissue around 3.24 kPa [4]; dynamic FE setting of 10 kPa [18, 36].	Fibroglandular tissue is commonly treated as stiffer or more structurally complex than adipose tissue. The highest reported values should be interpreted as method-dependent extremes, not typical values.	0.08–272 kPa
Skin layer	Reported skin values span from very soft effective model values to MPa-level stiffness assumptions, including 0.088–3000 kPa in the range reported by Hsu et al. [17], 22.72 kPa in patient-specific calibration [19], and 500 kPa in a dynamic multi-component FE model [36].	Skin stiffness is highly uncertain and strongly affects the model response, so it was treated as a sensitivity parameter.	0.088–3000 kPa
Tumor tissue	Tumor and malignant-mass stiffness values are generally higher than normal breast tissue, but reported values depend strongly on modality and region of interest. Samani et al. reported high-grade IDC around 42.52 ± 12.47 kPa [4], McKnight et al. reported carcinoma MRE regions around 18–94 kPa [42], while shear-wave elastography studies reported malignant breast masses around 114.9 ± 40.6 kPa [39] and breast-cancer stiffness values around $E_{\mathrm{mean}} = 131.17 \pm 52.76$ kPa and $E_{\max} = 169 \pm 70.3$ kPa [40].	Tumor stiffness can range from tens of kPa to above 100 kPa depending on measurement method. The 100 kPa tumor-overlay setting was therefore used as a strong stiffness-contrast value, not as patient-specific tumor calibration.	18–169 kPa

For comparison with these literature values, the main COMSOL material settings are summarised in Table 15. For the nearly incompressible two-coefficient Mooney–Rivlin route used in this project, the approximate small-strain stiffness scale was estimated as $E \approx 6(C_{10} + C_{01})$. This estimate was used only to compare stiffness scales; it does not replace the hyperelastic material definition.

Table 15: Approximate stiffness scales used in the current COMSOL model and sensitivity cases.

Component	Setting	Input definition	Approximate stiffness scale
Adipose tissue	Reference	$K = 425$ kPa, $C_{10} = 310$ Pa, $C_{01} = 300$ Pa	$E \approx 3.66$ kPa
	Soft sensitivity	$K = 425$ kPa, $C_{10} = 109$ Pa, $C_{01} = 106$ Pa	$E \approx 1.29$ kPa
Fibroglandular tissue	Reference	$K = 425$ kPa, $C_{10} = 833$ Pa, $C_{01} = 833$ Pa	$E \approx 10.0$ kPa
	Soft sensitivity	$K = 425$ kPa, $C_{10} = 230$ Pa, $C_{01} = 195$ Pa	$E \approx 2.55$ kPa
	Soft sensitivity	$K = 480$ kPa, $C_{10} = 1200$ Pa, $C_{01} = 1200$ Pa	$E \approx 14.4$ kPa
Skin	Intermediate sensitivity	$K = 8.33$ MPa, $C_{10} = 7333$ Pa, $C_{01} = 7333$ Pa	$E \approx 88$ kPa
	Stiff setting	$K = 8.33$ MPa, $C_{10} = 41667$ Pa, $C_{01} = 41667$ Pa	$E \approx 500$ kPa
Posterior support	Support surrogate	Linear elastic material with $E = 10$ kPa and $\nu = 0.49$	$E = 10$ kPa
Tumor overlay	Hard stiffness-contrast setting	$K = 425$ kPa, $C_{10} = 8333$ Pa, $C_{01} = 8333$ Pa	$E \approx 100$ kPa

Overall, the selected parameters should be interpreted as a controlled modelling set for comparing material routes and sensitivity cases. The adipose and fibroglandular reference values remain within the low-kPa soft-tissue scale reported in several breast-modelling studies. The skin and tumor settings cover a broader range because these components were used to test the mechanical influence of a stiff outer layer and a local stiffness inclusion. The values therefore define consistent stiffness scales for the COMSOL workflow, while acknowledging that breast tissue, skin and tumor properties cannot be represented by a single universally accepted parameter value.

F Additional result figures

This appendix provides additional time-dependent result figures that support the main Results chapter. These plots are included as supplementary material because the main Results section focuses on compact tables and selected peak-response figures. The additional figures show examples of the broader post-processing output that can be generated from the COMSOL pipeline.

For the time-dependent plots in this appendix, the horizontal axis represents the COMSOL simulation time. From 0 to 1.0 s, gravity is ramped to obtain the preloaded state. From 1.0 to 1.25 s, the model remains in the gravity-loaded equilibrium state before the dynamic input is applied. The additional acceleration pulse or prescribed support motion is then applied from 1.25 to 1.85 s, corresponding to the 0.60 s excitation duration used in the main dynamic cases. The remaining interval from 1.85 to 2.20 s shows the post-excitation response of the model.

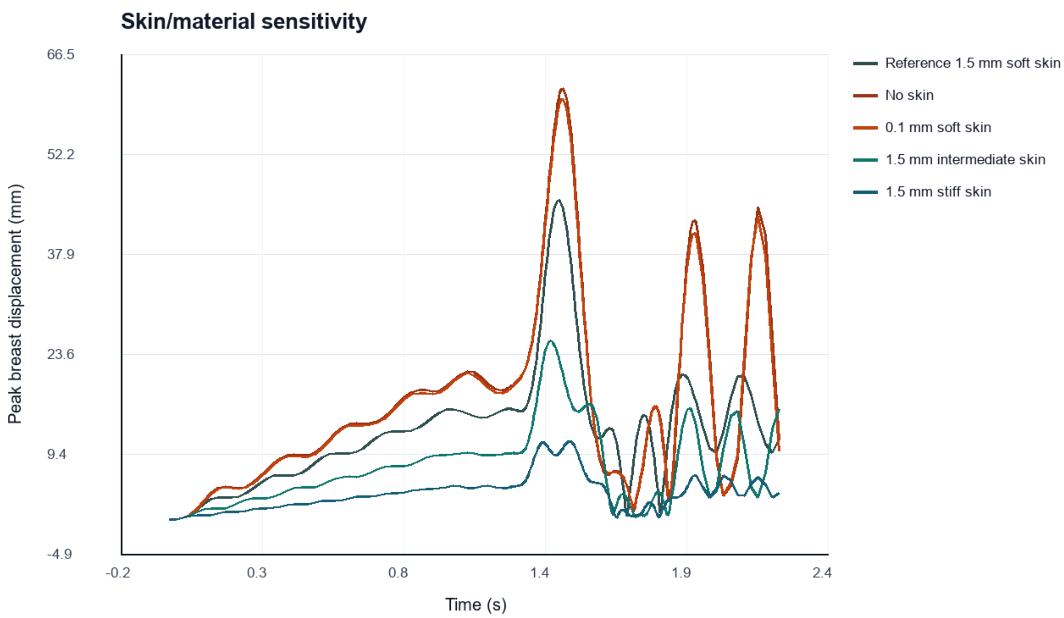

Figure 11: Time-dependent peak breast displacement for the simple-gland skin/material sensitivity cases. The no-skin and thin soft-skin cases show larger displacement amplitudes, while the stiffer volumetric-skin variants show reduced displacement.

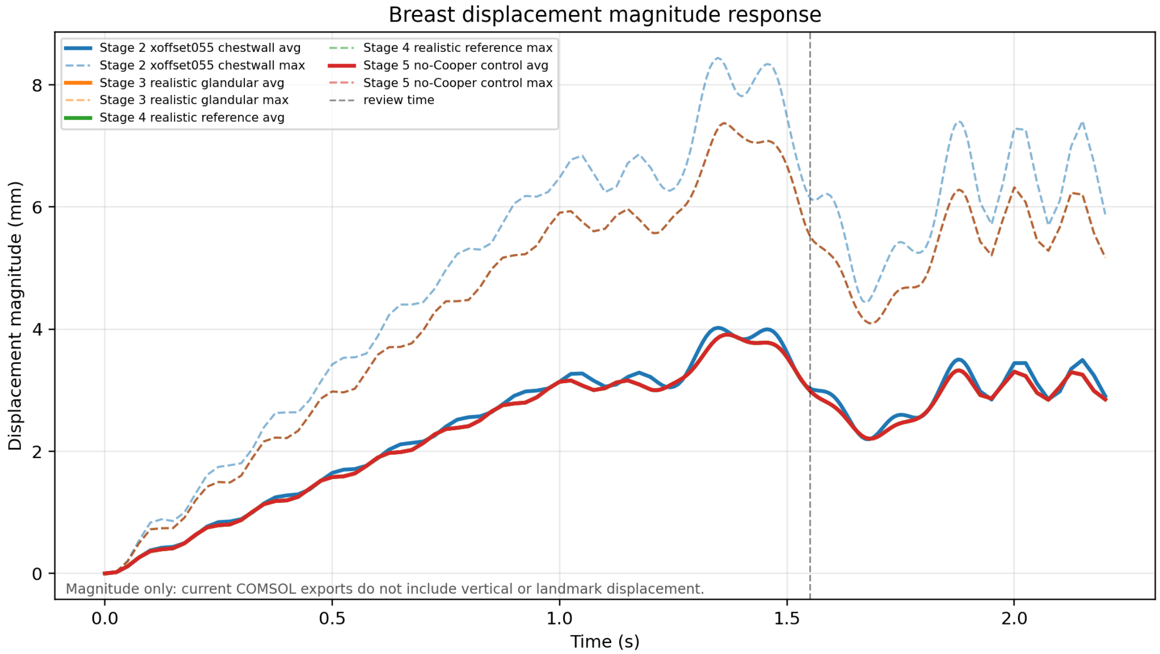

Figure 12: Time-dependent displacement-magnitude response for selected model-development stages. The figure is included as a development-stage comparison and shows how the response changed after introducing the curved chestwall route, realistic glandular distribution and no-Cooper Stage 5 reference case.

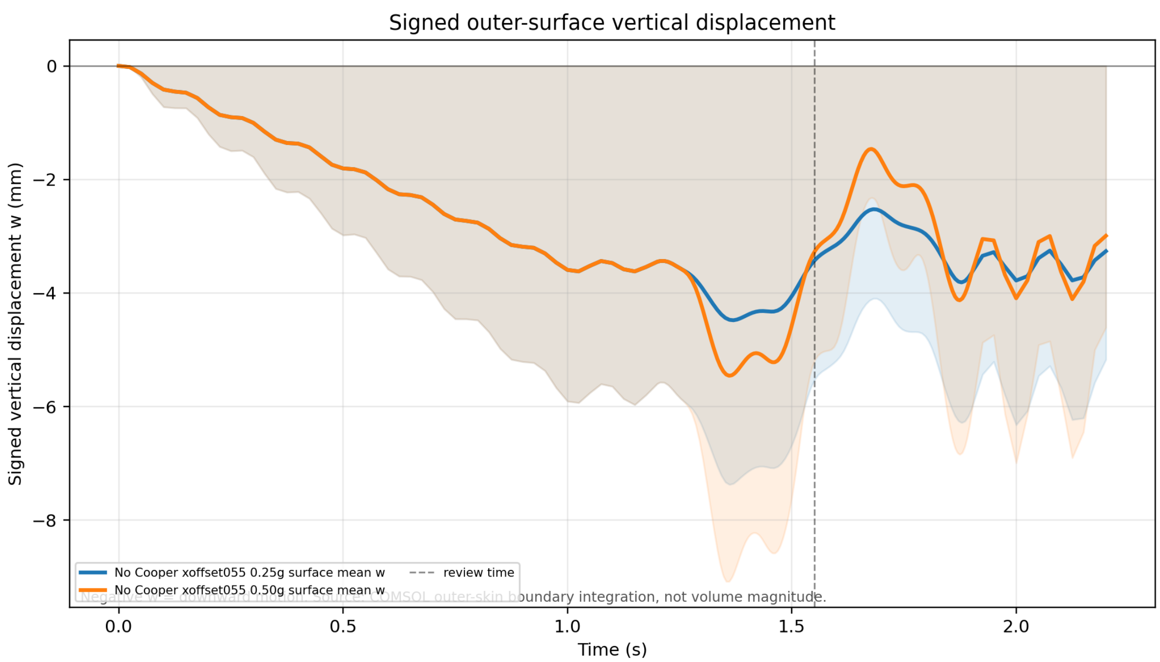

Figure 13: Time-dependent signed vertical surface displacement for the fixed-support acceleration-amplitude scout. The plot illustrates the surface-motion response for the $0.25g$ and $0.50g$ cases and supports the dynamic-loading results reported in Section 4.6.